Section 26 05 13 - Power Plant Wire and Cable (Combined Cycle) - Guide Specification

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Guide specification for wire and cable in a gas-fired combined cycle power plant (1x1, 2x1 or 3x1), written to US practice (NEC, ICEA, IEEE, UL). Text in [brackets] is an option or value for the project editor to confirm.

Part 1 - General

1.1 Summary

1. This section covers furnishing, installing, terminating and testing all field wire and cable for the plant, from the generator terminals to the switchyard and from the auxiliary switchgear to every load.
2. Included: 15 kV power cable, 600 V power and VFD cable, DC and UPS cable, control, instrument, fiber optic and grounding conductors, connectors, terminations and identification.
3. Not included: isolated phase and nonsegregated phase bus, 600 V busway, and overhead switchyard and line conductors (specified in their own sections); cable supplied inside OEM packages, except where it connects to field cable.

1.2 References

Use the edition current at contract award unless noted.

Body	Documents
NFPA	NFPA 70 (NEC) [2023]: Articles 110, 250, 310, 311, 330, 336, 392, 430, 445, 500-504, 725, 727, 770; Chapter 9 Tables 8 and 9
ICEA / NEMA	S-93-639 (WC74) 5-46 kV shielded power cable; S-95-658 (WC70) 600 V power cable; S-73-532 (WC57) control, thermocouple extension and instrumentation cable; P-32-382 short-circuit characteristics

Body	Documents
UL	44, 1072, 1277, 1569, 1685, 2225, 2250, 1424, 486A-486B, 467
IEEE	422, 1185, 525, 835, 1202, 400, 400.2, 404, 80, 81
NETA	ATS, Acceptance Testing Specifications
Owner	[Owner electrical design criteria]; [Transmission owner interconnection standards]; [Site hazardous area classification drawings]

1.3 Definitions

- **MV:** medium voltage, 2.4-34.5 kV; here 13.8 kV systems on 15 kV cable.
- **LV:** 600 V class and below.
- **MC-HL:** Type MC cable listed for hazardous locations, continuous corrugated welded armor with overall jacket.
- **TC-ER:** tray cable listed for exposed runs.
- **PLTC / ITC-HL:** power-limited tray cable / instrumentation tray cable for hazardous locations.
- **CI:** circuit integrity (fire-rated) cable.
- **IS:** intrinsically safe circuit.

1.4 System description

System	Nominal voltage	Grounding	Cable class
Generator	[18-24] kV	High-resistance via NGR	Phase bus (not this section); neutral cable 15 kV
Auxiliary MV	[13.8] kV [and 4.16 kV], 3-ph, 60 Hz	[Low-resistance, 400 A]	15 kV MV-105, 133%
Auxiliary LV	480 V, 3-ph, 3-wire	[High-resistance]	600 V
DC	125 V DC	Ungrounded	600 V

System	Nominal voltage	Grounding	Cable class
UPS / instrument AC	120/208 V	Solidly grounded	600 V
Instrument	24 V DC, 4-20 mA, thermocouple	Single-point shield ground	300 V

1.5 Submittals

1. Manufacturer data sheets for each cable type: construction, OD, mass, ampacity basis, short-circuit rating, minimum bend radius, maximum pulling tension and sidewall pressure.
2. Cable schedule: cable number, from/to equipment, type, size, conductors, runs, raceway route, length, design current, ampacity, voltage drop, short-circuit check, connector at each end.
3. Pulling calculations for every MV cable and every LV pull longer than [150 ft] or with more than [270°] of bends.
4. Certified factory test reports for each reel.
5. Field test reports and as-built cable schedule.

1.6 Quality assurance

1. Manufacturer: at least [10] years producing the cable types offered, ISO 9001 certified.
2. All cable of one type from one manufacturer.
3. Field testing by a NETA-accredited agency.
4. Terminators of 15 kV cable certified by the termination-kit manufacturer.

1.7 Delivery, storage and handling

1. Ship on reels with sealed ends, reel number, type, size and length marked.
2. Store reels on hard ground, ends sealed, protected from sunlight and damage.
3. Do not install below the manufacturer's minimum installation temperature [-10 °C / 14 °F for PVC jackets] without warming.

Part 2 - Products

2.1 General requirements (all cable)

1. Conductors: annealed copper, Class B compressed or Class C stranded; tinned where noted. No aluminum conductors.
2. Listed and marked for the installation: tray-rated (-TC or -ER), sunlight resistant, rated for direct burial where installed in duct banks.
3. Flame test: IEEE 1202 / UL 1685 FT4.
4. Construction policy: [Type MC-HL with continuous corrugated welded aluminum armor (CCW) and PVC jacket throughout] OR [TC-ER in tray; MC-HL in hazardous areas, exposed drops and where mechanical protection is needed].
5. Jacket: [PVC] OR [low-smoke zero-halogen in enclosed buildings and control rooms].
6. Minimum temperature rating: 90 °C wet and dry; 130 °C emergency; 250 °C short circuit (600 V XLPE). MV-105: 105 °C normal.

2.2 15 kV medium-voltage power cable

Item	Requirement
Standard	ICEA S-93-639, UL 1072, Type MV-105
Conductor	Copper, Class B compressed; #2 AWG minimum
Insulation	EPR, 133% insulation level, 220 mil nominal
Shields	Extruded semiconducting conductor and insulation shields; copper tape shield, 12.5% overlap
Configuration	[3/C with ground] for feeders up to [750 kcmil]; [1/C triplexed] above
Armor and jacket	[CCW aluminum armor, MC-HL] + PVC jacket, red for MV identification
Hazardous locations	Listed Class I Div 1 & 2 where routed through classified areas
Short circuit	Conductor and shield sized for the available fault current and clearing time per ICEA P-32-382

2.3 600 V power cable

Item	Requirement
Standard	ICEA S-95-658, UL 44 / UL 1569 (MC) or UL 1277 (TC)
Conductor	Copper; #12 AWG minimum for power
Insulation	XLPE, Type XHHW-2, 90 °C wet and dry
Configuration	3/C + insulated or bare ground for 3-phase; 2/C + ground for single-phase and DC
Armor and jacket	Per 2.1 item 4
Parallel runs	1/0 AWG minimum, equal length, same type (NEC 310.10(G))

2.4 VFD cable

1. Motor (output) side of every variable frequency drive.
2. Construction: 3/C copper, XLPE (XHHW-2), three symmetrical ground conductors, overall shield (copper tape or foil plus braid), [1000] V rated.
3. Fixed runs in tray: armored, Type MC-HL [Armor-X VFD type].
4. Final drops to vibrating motors (ACC fans, cooling tower fans): flexible tinned copper, Type TC-ER, NFPA 79 [Machine Flex type]. Not permitted in Class I Div 1.
5. MV drives: 15 kV shielded cable per 2.2, rated for the drive's dv/dt and common-mode voltage as confirmed by the drive manufacturer.
6. Cable length from drive to motor within the drive manufacturer's limit, or provide output filters.

2.5 DC and UPS cable

1. 600 V XHHW-2 copper, 2/C + ground, or 1/C pairs for large DC feeders; armor per 2.1 item 4.
2. Emergency lube oil pumps, emergency lighting and other circuits that must run during a fire: circuit integrity cable, [2-hour] rated (UL 2196), or protected routing.

2.6 Control cable

Item	Requirement
Standard	ICEA S-73-532 / S-95-658, UL 1277 or UL 1569
Insulation	XLPE, XHHW-2, 600 V
Conductor size	#14 AWG minimum; #12 AWG for 120 V AC power to panels; #10 AWG for CT secondaries
Conductor count	Standard counts [2, 3, 4, 7, 12, 19, 25, 37]; [20%] spare conductors, minimum 2
Color code	ICEA Method 1, Table E-2 (or printed numbers)

2.7 Instrument and thermocouple cable

Item	Requirement
Standard	ICEA S-73-532; NEC 725 (PLTC) / 727 (ITC)
Construction	Twisted pairs or triads, #16 AWG single pair, #18 AWG multi-pair; individual and overall aluminum-polyester shields with drain wire
Rating	300 V [600 V where run with power-limited 120 V circuits]
Armor	[CCW armor, PLTC-ER / ITC-HL]
Thermocouple	Extension grade matching the element (Type KX for K, JX for J), ANSI color code
Intrinsically safe circuits	Light blue jacket, separate from non-IS circuits (NEC 504.30)
Fieldbus / serial	Cable to the protocol's specification (RS-485 120 Ω, Foundation Fieldbus Type A)
Spares	[20%] spare pairs on multi-pair cable

2.8 Fiber optic cable

1. Single-mode OS2, [24 / 48] fibers, loose tube, gel-free.
2. Armored [interlocked aluminum] OFCR or OFNR; OPGW on transmission line and switchyard shield wire where specified.
3. Pre-terminated or fusion-spliced; [25%] spare fibers.

2.9 Grounding conductors

1. Ground grid: bare soft-drawn stranded copper, [4/0 AWG] minimum or as the IEEE 80 study requires.
2. Equipment bonding jumpers: copper, [2/0 AWG] minimum for major equipment; green insulated where exposed.
3. Connections below grade: exothermic weld or compression listed to IEEE 837.

2.10 Connectors, terminations and splices

1. MC and MC-HL connectors: listed for the cable OD and armor type; watertight; [stainless or aluminum].
2. Hazardous locations: connectors listed to UL 2225 for the Class and Division; seals per NEC 501.15 in Division 1.
3. VFD cable: EMC connectors with 360° shield termination at both ends.
4. 15 kV terminations: IEEE 48 Class 1, cold-shrink or heat-shrink kits, [outdoor / indoor]; shield grounding per manufacturer.
5. 15 kV splices: avoid; where unavoidable, IEEE 404 rated and recorded on the as-built schedule.
6. Lugs: compression, tin-plated copper, two-hole for 1/0 AWG and larger.

2.11 Identification

1. Cable tag at both ends and at every pull point and tray entry: cable number per the cable schedule, on [stainless steel / UV-stable] tags.
2. Conductor markers on every conductor at terminations: terminal number per wiring diagram.
3. MV cable: phase identification tape and "DANGER - 13.8 kV" markers at accessible locations.

Part 3 - Execution

3.1 Cable sizing criteria

Every power cable shall meet all four checks below; the cable schedule shows each result.

Check	Criterion
Ampacity	Design current ⩽ allowable ampacity. Design current = 125% of motor full-load or continuous load (NEC 430.22, 215.2); 115% of generator rating (NEC 445.13); 100% of transformer secondary. Allowable = NEC 310.16 75 °C column (600 V) or NEC 311.60 (15 kV), corrected for [40 °C] ambient and installation (NEC 392.80 for tray; IEEE 835 / Neher-McGrath, RHO [90], for duct banks).
Voltage drop	⩽ [3%] feeder and ⩽ [5%] feeder plus branch at full load; motor terminal voltage ⩾ [80%] during starting. NEC Chapter 9 Table 9 impedances at the load power factor.
Short circuit	Conductor and shield withstand the maximum available fault for the protective device clearing time (ICEA P-32-382). Governs most 13.8 kV feeders on multi-GT plants.
Minimum size	15 kV: #2 AWG. 600 V power: #12 AWG. Control: #14 AWG. CT circuits: #10 AWG.

3.2 Raceway and segregation

1. Cable tray: ladder type; fill per NEC 392.22 (cables 4/0 AWG and larger in a single layer); [25%] spare capacity at turnover.
2. Separate trays or tray tiers by level, top to bottom: 15 kV; 600 V power; control; instrument. Instrument tray at least [12 in] from 600 V power tray, or a solid barrier.
3. Duct banks: concrete-encased, [red-dyed]; [25%] spare ducts; ampacity per IEEE 835.
4. Unit separation: each generating unit (each GT/HRSG, ST) on its own tray routes where practical, so a cable fire affects one unit only. Redundant trains (A/B pumps, redundant controllers, fiber pairs) on separate routes (IEEE 422).

5. Switchyard control cable in trench or duct to the relay building per IEEE 525 and transmission owner standards.

3.3 Installation

1. Follow IEEE 1185 and the manufacturer's limits for pulling tension, sidewall pressure and minimum bending radius. Use a dynamometer on all MV pulls and record the maximum tension.
2. Use pulling lubricant compatible with the jacket. Pull from reels with the cable entering the raceway without twisting.
3. Secure cable in vertical tray and at changes of direction; support armored cable per NEC 330.30.
4. Train cables neatly; maintain bend radius at terminations. Single-conductor MV in triplexed or trefoil groups with all phases in the same raceway.
5. Seal cable ends until terminated. Fire-stop all wall and floor penetrations with a listed system.

3.4 Hazardous locations

1. Install per NEC 501 and the site hazardous area classification drawings.
2. Class I Division 1 (LNG pump and tank-top penetrations, gas metering where classified Div 1): MC-HL or ITC-HL with UL 2225 Division 1 connectors; seals within 18 in of enclosures per NEC 501.15.
3. Class I Division 2 (gas metering skids, fuel gas heaters and compressors, GT enclosures, BOG compressors): MC-HL, ITC-HL, TC-ER or PLTC with listed Division 2 fittings.
4. Intrinsically safe circuits: separate raceway or 2 in separation, light blue, per the control drawing.

3.5 Terminations

1. Terminate per manufacturer instructions with calibrated tools; torque to the equipment rating and mark.
2. Ground MV cable shields at [both ends] (single-point grounding only where approved for long single-conductor runs).
3. Ground instrument shields at the [control system] end only; insulate the field end.
4. Ground VFD cable shields and symmetrical grounds at both ends.

3.6 Field testing

Test every cable after installation and before energizing, per NETA ATS. Values in brackets are typical; confirm against NETA ATS and the cable manufacturer.

Cable	Test	Acceptance
All	Visual and mechanical: damage, bend radius, tags, connector tightness	No defects
600 V power, control, DC	Insulation resistance, [1,000 V DC], 1 min, conductor to conductor and to ground	≥ [100 MΩ], or per NETA ATS Table 100.1
600 V power, control	Continuity and point-to-point against wiring diagrams	100% correct
15 kV	Insulation resistance, [5,000 V DC], before and after withstand	Per NETA ATS; no decrease after withstand
15 kV	Shield continuity	Continuous end to end
15 kV, splices, terminations	VLF withstand, 0.1 Hz sinusoidal, IEEE 400.2 acceptance level [21 kV rms for 15 kV cable], [60 min]	No breakdown
15 kV	VLF tan δ (IEEE 400.2)	Within IEEE 400.2 acceptance criteria for EPR; record baseline
VFD	Insulation resistance, continuity of symmetrical grounds and shield	As 600 V
Instrument	Loop check from field device to control system; shield isolation from ground at field end	100% correct
Fiber	OTDR both directions and end-to-end insertion loss	Within the link loss budget
Grounding	Fall-of-potential grid resistance (IEEE 81); continuity of equipment bonds	≤ [1 Ω] grid resistance or per the IEEE 80 study

Do not use DC hi-pot on aged or service-aged 15 kV cable.

3.7 Records

1. Maintain the cable schedule as the master record. For each cable, record reel number, date pulled, maximum pull tension (MV), date terminated at each end and test results.
2. Hand over: as-built cable schedule (native spreadsheet), test reports, pulling records, termination records and manufacturer data sheets.
3. A cable is complete when pulled, terminated at both ends, tested and accepted. Report status weekly by system.

Appendix A - Cable type schedule

Every field cable in the plant shall be one of the types below. The cable schedule references these codes; add a type only by revising this appendix. Code format: `class-conductors-size`; suffix `-A` = armored (CCW, MC-HL / ITC-HL), `-T` = unarmored tray cable (TC-ER / PLTC-ER), applied per 2.1 item 4.

A.1 15 kV medium-voltage power (2.2)

Common construction: Cu Class B compressed; EPR 220 mil, 133%; semiconducting shields; Cu tape shield; overall PVC jacket (red); MV-105, UL 1072, ICEA S-93-639; armored versions MC-HL listed Class I Div 1 & 2.

Code	Conductors	Sizes (AWG / kcmil)	Typical use
MV15-1C-[size]-A	1/C	#2	Generator neutral to NGR
MV15-3C-[size]-A	3/C + GC	#2, 1/0, 2/0, 4/0, 250, 350, 500, 750	Motor and transformer feeders (#2 min: pumps, fans; 4/0 BFPs; 500-750 large fans, compressors, ties)
MV15-1C-[size]-TPX-A	3 × 1/C triplexed	500, 750, 1000	Feeders above 3/C range: switchgear incomers, RICE / CCS bus ties where busduct is not used

Code	Conductors	Sizes (AWG / kcmil)	Typical use
MV15-3C-[size]-VFD-A	3/C + 3 symmetrical G	2/0 - 750	MV drive output (e.g. CO_2 compressor), subject to drive maker's dv/dt rating

A.2 600 V power (2.3)

Common construction: Cu Class B; XLPE XHHW-2, 90 °C wet/dry; 3/C + insulated green ground (sized per NEC 250.122) unless noted; PVC jacket (black); UL 1569 MC-HL (-A) or UL 1277 TC-ER (-T); FT4.

Code	Conductors	Sizes (AWG / kcmil)	Typical use
LV-3C-[size]-A/T	3/C + G	#12, #10, #8, #6, #4, #2, 1/0, 2/0, 4/0, 250, 350, 500	480 V motor, heater, MCC and panel feeders
LV-4C-[size]-A/T	4/C + G	#12 - 4/0	208Y/120 V and 480Y/277 V 4-wire feeders (lighting, small power panels)
LV-1C-[size]-A	1/C	250 - 750	Large parallel 480 V feeders where 3/C cannot be used
LV-VFD-[size]-A	3/C + 3 symmetrical G, overall shield	#12 - 500	480 V VFD output, fixed tray runs (Armor-X VFD type)
LV-MFX-[size]-T	3/C + 3 symmetrical G, flexible tinned Cu, foil + braid shield, TPE jacket, 1000 V, TC-ER, NFPA 79	#14 - [2/0, confirm range]	Final drop to vibrating VFD motors (ACC and cooling-tower fans); not Class I Div 1 (Machine Flex type)

A.3 DC and UPS power (2.5)

Code	Construction	Sizes	Typical use
DC-2C-[size]-A	2/C + G, XHHW-2, 600 V	#12 - 4/0	125 V DC distribution, trip and close supplies; 120/208 V UPS feeders
DC-1C-[size]-A	1/C pairs, XHHW-2, 600 V	250 - 500	Battery to DC switchboard; large DC motors (emergency lube oil pumps)
CI-2C-[size]	2/C + G, circuit integrity, UL 2196 [2-hour]	#12 - 4/0	Emergency lube oil pumps, emergency lighting, fire & gas power where routed through fire zones

A.4 Control (2.6)

Common construction: Cu Class B (Class C / tinned optional); XLPE XHHW-2, 600 V; ICEA Method 1 color code; PVC jacket (black); -A or -T; FT4.

Code	Conductors	Size	Typical use
CTL-[n]C-14-A/T	2, 3, 4, 7, 12, 19, 25, 37	#14	General control, interlocks, status and alarm wiring
CTL-[n]C-12-A/T	2, 3, 4, 7, 12	#12	120 V AC control power, space heaters, solenoids, MOV controls
CTL-[n]C-10-A/T	4, 7, 12	#10	CT secondaries, VT circuits, long DC trip circuits
CTL-[n]C-14-S-A	4, 7, 12	#14, overall shield	Control near VFDs or in high-noise areas (synchronizing, speed signals)

A.5 Instrument and thermocouple (2.7)

Common construction: Cu Class B; PE or FR-PVC insulation; twisted, individually and overall shielded (Al-polyester + drain); 300 V [600 V option]; PLTC-ER / ITC-ER (-T) or ITC-HL armored (-A); black jacket, light blue for IS.

Code	Pairs / triads	Size	Typical use
INS-1P-16-A/T	1 pair	#16	Single 4-20 mA / HART field device home run
INS-1T-16-A/T	1 triad	#16	RTDs (3-wire), 3-wire transmitters
INS-[n]P-18-A/T	4, 8, 12, 24, 36 pairs	#18	Junction box to control system multi-pair
INS-[n]T-18-A/T	4, 8, 12 triads	#18	Multi-RTD runs (bearing, winding temperatures)
INS-IS-[n]P-[size]	As above	#16 / #18	Intrinsically safe circuits, light blue jacket
TC-KX-[n]P-16-A/T	1, 4, 8, 12 pairs	#16	Type K thermocouples (GT exhaust, HRSG tubes, bearings); ANSI color code
TC-JX-[n]P-16-A/T	1, 4, 8 pairs	#16	Type J thermocouples where used

A.6 Communication and data

Code	Construction	Typical use
COM-RS485-1P-22	1 TSP #22-24 AWG, 120 Ω, shielded, ITC-ER / -HL	Modbus RTU: flow computers, analyzers, meters
COM-FF-A	Foundation Fieldbus Type A, 1 TSP #18 AWG, 100 Ω	Fieldbus segments
COM-PB-DP	PROFIBUS DP, 1 pair #22 AWG, 150 Ω	PROFIBUS networks (if used by OEM packages)
COM-CAT6A-S	Shielded Cat 6A, outdoor / tray-rated	Ethernet inside buildings and e-houses (≤ 90 m)

A.7 Fiber optic (2.8)

Code	Construction	Typical use
FO-[12/24/48]SM-A	Single-mode OS2, loose tube, gel-free, interlocked armor, OFCR	DCS backbone, protection intertrip, switchyard SCADA, CCTV / security
FO-[12/24]MM-A	Multimode OM4, armored	Short runs inside buildings where equipment requires multimode
OPGW-[24/48]	Optical ground wire	Transmission line and switchyard shield wire with teleprotection fibers

A.8 Grounding (2.9)

Code	Construction	Typical use
GND-BARE-[4/0, 500]	Bare soft-drawn stranded Cu	Buried ground grid and ring grounds per IEEE 80 study
GND-BARE-2/0	Bare stranded Cu	Grid risers to structures, fences, tanks, lightning down conductors
GND-GRN-[#6 - 4/0]	Green insulated stranded Cu, THHN/THWN-2 or XHHW-2	Equipment bonding jumpers above grade, panel and tray bonding
GND-TRAY-[#4/0 bare or 2/0 insulated]	Cu	Tray ground conductor run full length of power tray where used as EGC path

Size ranges are typical. Confirm each type's available sizes, OD, mass and ratings against the selected manufacturer's data sheets before procurement.

Editor's notes

Fill every bracketed value from the inputs below before issue. Owner and utility requirements override this guide.

- [ ] Plant configuration (1x1, 2x1, 3x1) and one-line diagram: sets step-up transformers, switchyard bays and unit separation (3.2 item 4).
- [ ] Owner electrical design criteria: construction policy (2.1 item 4), jacket type, insulation, spare conductors and spare tray / duct capacity.
- [ ] Auxiliary system grounding (1.4) and short-circuit study: fault levels and clearing times for 3.1.
- [ ] Transmission owner standards: switchyard control cable, CT/VT wiring, fiber and OPGW (2.8, 3.2 item 5).
- [ ] Site hazardous area classification drawings (3.4).
- [ ] OEM interface lists for GT, ST, HRSG, CCS and drives: scope split between vendor and field cable; drive cable limits (2.4).
- [ ] Load list: motor ratings, starting method (across-the-line or VFD), duty.
- [ ] Test acceptance values: confirm NETA ATS edition and IEEE 400.2 levels (3.6).

Basis of this guide. It follows the NEC, ICEA, UL and IEEE documents in 1.2, and the structure of UFGS 26 05 13 (Medium-Voltage Cables) and UFGS 26 05 19.00 10 (Insulated Wire and Cable). Product requirements align with published listings for armored MC-HL power, control and VFD cable, 15 kV MV-105 cable and flexible TC-ER VFD cable. These sources were identified by search but not read in full; bracketed values are typical practice. See the Gas & Combined Cycle Plant Cable Requirements reference for the source list.